

applications, and is the integration of IoT with the cloud and the World Wide Web for the real-time connectivity of devices.

A number of programming platforms are available for working with IoT, IoE, fog or edge scenarios. Hundreds of toolkits, which are very powerful and easy to use for dynamic research, are available as well. A few of these are listed in Table 1.

Node-RED: A tool for flow based programming of IoT scenarios

Node-RED (<https://nodered.org/>) is a powerful and easy-to-use programming platform for the simulation of IoT scenarios. Fog and edge computing can also be done using flow based programming in Node-RED. Here, high performance structures can be implemented using minimum coding.

Installation and working

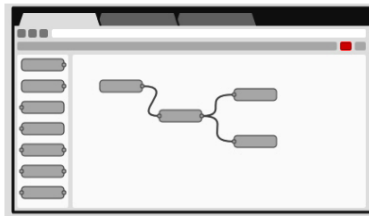
Node-RED is a specialised package that is installed onto the Node.js platform. The latter is a lightweight but high performance programming environment based on JavaScript. Many packages are available in Node.js for multiple applications, including Internet of Things (IoT), cloud computing, machine learning, data science, and blockchain.

To work with Node-RED, the Node.js platform should be installed first, which is available at <https://nodejs.org> for multiple operating systems like Windows, Mac and Linux for 32-bit or 64-bit architectures (Figure 1).

After the installation of Node.js, the Node-RED package can be installed from the node package manager (NPM), which is the repository of packages developed and deployed for the Node platform (Figure 2).

In the installation directory of Node.js, the `npm` command is used to install Node-RED, as follows:

```
E:\>cd nodejs
E:\nodejs>npm i node-red
```



Browser-based flow editing

Node-RED provides a browser-based flow editor that makes it easy to wire together flows using the wide range of nodes in the palette. Flows can be then deployed to the runtime in a single-click.

JavaScript functions can be created within the editor using a rich text editor.

A built-in library allows you to save useful functions, templates or flows for re-use.

Built on Node.js

The light-weight runtime is built on Node.js, taking full advantage of its event-driven, non-blocking model. This makes it ideal to run at the edge of the network on low-cost hardware such as the Raspberry Pi as well as in the cloud.

With over 225,000 modules in Node's package repository, it is easy to extend the range of palette nodes to add new capabilities.

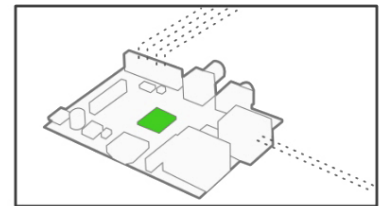


Figure 2: Node-RED platform built on Node.js (Source: nodered.org)

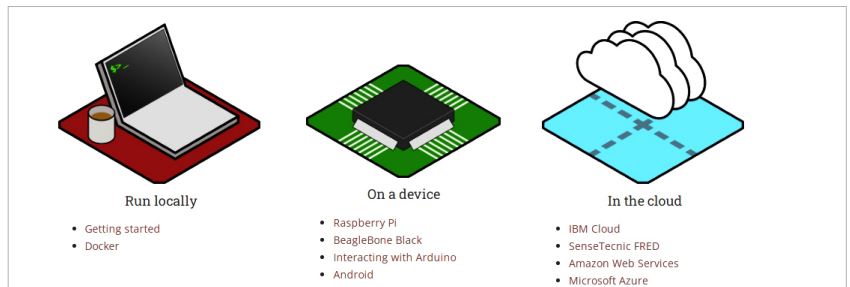


Figure 3: Installation options for Node-RED (Source: nodered.org)

```
node-red
E:\nodejs>node-red
21 Nov 09:55:07 - [info]
Welcome to Node-RED
=====
21 Nov 09:55:07 - [info] Node-RED version: v1.0.4
21 Nov 09:55:07 - [info] Node.js version: v12.16.1
21 Nov 09:55:07 - [info] Windows_NT 6.1.7600 x64 LE
21 Nov 09:55:15 - [info] Loading palette nodes
21 Nov 09:56:27 - [warn] [node-red-contrib-noble/noble] Error: Cannot find module
'bluetooth-hci-socket'
Require stack:
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\lib\hci-socket\hci.js
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\lib\hci-socket\bindings.js
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\lib\resolve-bindings.js
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\index.js
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\node-red-contrib-noble.js
- E:\nodejs\node_modules\node-red\node_modules\node-red\registry\lib\loader.js
- E:\nodejs\node_modules\node-red\node_modules\node-red\registry\lib\index.js
- E:\nodejs\node_modules\node-red\node_modules\node-red\runtime\lib\nodes\index.js
- E:\nodejs\node_modules\node-red\node_modules\node-red\runtime\lib\index.js
- E:\nodejs\node_modules\node-red\lib\red.js
- E:\nodejs\node_modules\node-red\red.js
21 Nov 09:56:27 - [warn]
21 Nov 09:56:27 - [info] Settings file : C:\Users\kumargaurav\.node-red\settings.js
21 Nov 09:56:27 - [info] Context store : 'default' [module=memory]
```

Figure 4: Starting the Node-RED server